

3.7 The Ratio Test

In previous sections we have seen that in order for a series $\sum_{n=1}^{\infty} a_n$ to converge it must be the case that $\lim_{n \rightarrow \infty} a_n = 0$. However, we also know that series like the harmonic series satisfy the condition that $\lim_{n \rightarrow \infty} a_n = 0$ but does not converge. So not only do we need $\lim_{n \rightarrow \infty} a_n = 0$, but we need to have $a_n \rightarrow 0$ quickly enough. In this section we develop a test to help us with this distinction.

Theorem (Ratio Test): Let $\sum_{k=1}^{\infty} a_k$ be a series with nonzero terms and $\rho = \lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|}$.

1. If $\rho < 1$, the series $\sum_{k=1}^{\infty} a_k$ converges absolutely and therefore converges.
2. If $\rho > 1$ or $\rho = +\infty$, the series $\sum_{k=1}^{\infty} a_k$ diverges.
3. If $\rho = 1$, the test is inconclusive and no conclusion can be drawn.

Proof: The proof is in the textbook.

Note: As we are used to, this theorem still holds if the index starts at a value other than 1.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{1}{k!}$ converges or diverges.

Example 2: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ converges or diverges.

Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^n 2^n}{n!} \right)$ is absolutely convergent, conditionally convergent, or divergent.

Example 4: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2 - 0.75}}$ is absolutely convergent, conditionally convergent, or divergent.